

Diary entry for Spectral-Sequences

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2026-06-30

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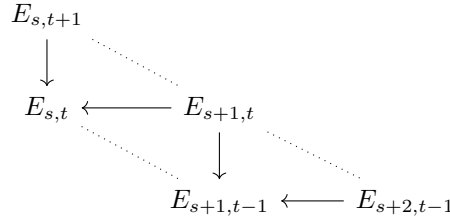
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1 Theory

Despite the name of the section, I will not be detailed about the mechanics of how to define particular differentials or how to justify convergence or anything of that sort. You can get a lot of juice out of just using the abundance of exact sequences produced by a spectral sequence!

1.1 Spectral sequence of a double complex

Definition 1.1 (Double complex, total complex). A homologically graded **double complex** of R -modules is a collection $E_{j,k}$ of R -modules along with maps $d^h: E_{j,k} \rightarrow E_{j-1,k}$ and $d^v: E_{j,k} \rightarrow E_{j,k-1}$ for each j,k such that $d^h \circ d^h = d^v \circ d^v = d^h \circ d^v + d^v \circ d^h = 0$. The **total complex** of a double complex E is the chain complex $T = \text{Tot}(E)$ whose n th module T_n is the product $\prod_{p+q=n} E_{p,q}$, with differential $d^T: T_n \rightarrow T_{n-1}$ given by setting the $E_{s,t}$ component of $d(x_{p,q})_{p+q=n}$ to be the sum $d^h x_{s+1,t} + d^v x_{s,t+1}$; see the diagram below, in which elements of the double complex on the same level are joined by dotted lines.



A cohomologically graded double complex is the obvious dual notion: flip the directions of the differentials so that they increase the degree, and the direction of differentials in the total complex should be adjusted as a result.

We will assume that all of our double complexes live in the first quadrant, which as we will see guarantees that our spectral sequence “converges” (in a sense that we will make precise).

Exercise 1.2. Verify that for a (co)homologically graded double complex E , the total complex $\text{Tot}(E)$ is indeed a (co)chain complex.

The spectral sequences of a double complex E are tools for computing the homology of the total complex of E . The idea is that one can approximate the homology by successively constructing a chain complex involving the various modules that $E_{s,t}$ “should be able to see” (namely the modules along the line $p + q = s + t - 1$). To this end, a spectral sequence for the double complex is defined as follows: one has a sequence of **pages** E^r , each of which contains a module for each pair of integers (s, t) , with $E^0_{s,t} = E_{s,t}$, and differentials d^r going out of each $E^r_{s,t}$, each shifting the degrees by the same amount, which partition the grid into chain complexes. Then, we stipulate that $E^{r+1}_{s,t}$ is the homology at (s, t) of the chain complex in page r which contains $E^r_{s,t}$.

picture

At this point we will make an arbitrary decision: suppose that the differentials in the E^0 page are the vertical differentials $d^0 = d^v$. It turns out that this provides us with naturally defined differentials $d^r: E^r_{s,t} \rightarrow E^r_{s-r,t+r-1}$; for example d^1 is the map on homology induced by the horizontal differentials d^h (note that each d^r decreases the total degree by 1, which is what we would like to see). Since we are assuming that $E_{s,t} = 0$ if s or t is negative and zeroes persist as we pass to later pages, the module at each spot (s, t) stabilizes as we go far into the sequence; we call the limiting page E^∞ .

words are hard (maybe define convergence earlier)

1.2 Spectral sequence of a filtered complex

TODO

(show the double complex thing as a special case of the filtered complex thing)

1.3 Spectral sequence of an exact couple

TODO

(WTF is an exact couple?)

2 Examples and applications

2.1 Some results in homological algebra

Proposition 2.1 (Four lemmas). Consider the following commutative diagram of modules (or just in some abelian category), in which the two rows are exact:

$$\begin{array}{ccccccc} A & \longleftarrow & B & \longleftarrow & C & \longleftarrow & D \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta \\ A' & \longleftarrow & B' & \longleftarrow & C' & \longleftarrow & D' \end{array}$$

- (a) If α and γ are monomorphisms and δ is an epimorphism, then β is an monomorphism.
- (b) If β and δ are epimorphisms and α is an monomorphism, then γ is a epimorphism.

(The diagram is drawn backwards to emphasize how the double complex is oriented in the plane.)

Proof. By extending via zeroes, we can view the given diagram as a first quadrant double complex E which is concentrated in finitely many points of the grid, so the corresponding (starting-at-horizontal or starting-at-vertical) spectral sequences converge to the homology of its total complex. If we take horizontal homology first to get to E^1 , then the E^1 (hence E^∞) page is all zeroes, so the spectral sequence must converge to zero.

Now we consider the two parts separately and pass to E^1 by taking vertical homology.

- (a) The E^1 page is

$$\begin{array}{ccccccc} 0 & \xleftarrow{d^1} & \ker \beta & \xleftarrow{d^1} & 0 & & \ker \delta \\ & & & \searrow d^2 & & & \\ \text{coker } \alpha & & \text{coker } \beta & & \text{coker } \gamma & & 0 \end{array}$$

From here on out, the differentials into and out of $\ker \beta$ are all zeroes: in E^1 , the differentials are the adjacent zero maps; in E^2 , the incoming differential is from $\text{coker } \delta = 0$ and the outgoing one goes outside the first quadrant; and in the remaining pages the differentials come from or end at a place outside the first quadrant. This means $\ker \beta$ persists in E^∞ , which we have already discovered consists of all zeroes, so $\ker \beta = 0$.

- (b) Similarly, the E^1 page in this case is

$$\begin{array}{ccccccc} 0 & & \ker \beta & & \ker \gamma & & \ker \delta \\ & \searrow d^2 & & & & & \\ \text{coker } \alpha & & 0 & \xleftarrow{d^1} & \text{coker } \gamma & \xleftarrow{d^1} & 0 \end{array}$$

and the differentials involving $\text{coker } \gamma$ from here on out are all zeroes, meaning $\text{coker } \gamma$ persists to E^∞ , hence is zero. $\square$

Exercise 2.2. Prove the five lemma using a similar spectral sequence argument. Recall that the five lemma says that for a commutative diagram

$$\begin{array}{ccccccccc} A & \longleftarrow & B & \longleftarrow & C & \longleftarrow & D & \longleftarrow & E \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta & & \downarrow \delta \\ A' & \longleftarrow & B' & \longleftarrow & C' & \longleftarrow & D' & \longleftarrow & E' \end{array}$$

in which the two rows are exact, β and δ are isomorphisms, δ is an epimorphism, and α is a monomorphism, we have that γ is an isomorphism. (In real life this often just manifests as the outer four maps being isomorphisms implying that γ is an isomorphism.)

Proposition 2.3 (Snake lemma). Consider the commutative diagram of modules

$$\begin{array}{ccccccc} 0 & \longleftarrow & C & \xleftarrow{g} & B & \xleftarrow{f} & A \\ & & \downarrow \gamma & & \downarrow \beta & & \downarrow \alpha \\ & & C' & \xleftarrow{g'} & B' & \xleftarrow{f'} & A' \longleftarrow 0 \end{array}$$

in which the rows are exact. There is an exact sequence

$$\ker f \rightarrow \ker \alpha \rightarrow \ker \beta \rightarrow \ker \gamma \xrightarrow{\partial} \operatorname{coker} \alpha \rightarrow \operatorname{coker} \beta \rightarrow \operatorname{coker} \gamma \rightarrow \operatorname{coker} g'.$$

Proof. As before, complete the diagram with zeroes to make it into a first quadrant double complex E . Taking horizontal homology gives an E^1 page of

$$\begin{array}{ccc} 0 & 0 & \ker f \\ \operatorname{coker} g' & 0 & 0 \end{array}$$

after which the spectral sequence stabilizes, so this is equal to the E^∞ page.

Taking vertical homology instead gives an E^1 page of

$$\begin{array}{ccc} \ker \gamma & \ker \beta & \ker \alpha \\ \operatorname{coker} \gamma & \operatorname{coker} \beta & \operatorname{coker} \alpha \end{array}$$

Passing to E^2 by taking horizontal homology must kill $\ker \beta$ and $\operatorname{coker} \beta$ because in future pages they are not involved in nonzero differentials. For a similar reason, the homology at $\ker \alpha$ must be $\ker f$ and the homology at $\operatorname{coker} \gamma$ must be $\operatorname{coker} g'$, since the objects at those locations also persist from E^2 to E^∞ . This gives the entire exact sequence except for the connecting homomorphism, which we consider now. In the E^2 page the only nontrivial differential is a map

$$d^2: \ker d_{2,0}^1 \rightarrow \operatorname{coker} d_{1,1}^1,$$

which must be an isomorphism because the objects at these locations must vanish in the E^∞ page and they will not be involved in nontrivial differentials in the future. Define the connecting map ∂ as the composition

$$\ker \gamma \twoheadrightarrow \operatorname{coker} d_{1,1}^1 \xrightarrow{\sim} \ker d_{2,0}^1 \hookrightarrow \operatorname{coker} \alpha,$$

inverting d^2 for the middle map. The kernel of ∂ is that of the first map in the composition, namely the image of the map $d_{1,1}^1: \ker \beta \rightarrow \ker \gamma$ in the exact sequence, so $\ker \beta \rightarrow \ker \gamma \xrightarrow{\partial} \operatorname{coker} \alpha$ is exact at $\ker \gamma$. Similarly, the image of ∂ is isomorphic to the kernel of the map in the exact sequence $d_{2,0}^1: \operatorname{coker} \beta \rightarrow \operatorname{coker} \alpha$, proving exactness at $\operatorname{coker} \alpha$. $\square$

Exercise 2.4. Use essentially the same argument to show that a short exact sequence of chain complexes induces a long exact sequence in homology.

Note that in all the examples above, the total complex itself is essentially irrelevant; the only important thing is that both spectral sequences furnished by the double complex converge to the same thing (which fortunately was all zeroes).

2.2 Serre spectral sequence

Theorem 2.5 (Serre spectral sequence). Let $F \hookrightarrow E \rightarrow B$ be a Serre fibration with path connected base space B whose fundamental group acts trivially on $H_*(F)$. Then, there is a homological spectral sequence $E_{p,q}^2 = H_p(B; H_q(F))$ converging to $H_*(E)$ and a cohomological spectral sequence $E_2^{p,q} = H^p(B; H^q(F; R))$ converging to $H^*(E)$ for some abelian group R . Furthermore, if R is a ring, there are bilinear products $E_r^{p,q} \otimes E_r^{s,t} \rightarrow E_r^{p+q, s+t}$ on each page such that

- (a) Each d_r is a derivation, namely $d_r(xy) = (d_r x)y + (-1)^{p+q} x d_r y$ where $x \in E_r^{p+q}$. Note that this means the product on page $r+1$ is induced by the product on page r .
- (b) The product in E^2 is the standard cup product with coefficients in the cohomology ring of the fiber.
- (c) The cup product on $H^*(E)$ induces the products on the E^∞ page.

This is particularly useful when you can say something nice about some of the terms of the spectral sequence, such as knowing that E is contractible or that there are a lot of zero rows/columns (perhaps the spectral sequence is concentrated in just a few rows or columns). Note also that the triviality of the action of π_1 often manifests as X just being simply connected. (To rid yourself of the triviality condition, one must work with “local coefficients,” which at the moment I know know nothing about.)

Here are a whole bunch of examples.

Theorem 2.6 (Hurewicz’s theorem). Suppose that X is n -connected, meaning the first n homotopy groups of X are zero. Then, the first n reduced homology groups with integer coefficients of X are also zero, and $\pi_{n+1}(X, x_0)^{\text{ab}} \cong H_{n+1}(X)$ (of course, the abelianization is superfluous for $n > 0$).

Proof. (This proof was shown to me by my friend Nick.)

For $n = 0$ one can prove the theorem manually without too much blood, sweat, or tears; see Hatcher appendix 2A. To extend to larger n , we induct on n . Suppose that X is n -connected for $n \geq 1$ (notably, simply connected), and consider the Serre spectral sequence for the path fibration $\Omega X \rightarrow PX \rightarrow X$. Using the inductive hypothesis and the fact that ΩX is $(n-1)$ -connected, the E^2 page of the Serre spectral sequence looks like

$$\begin{array}{cccccc}
 H_n(\Omega X) = \pi_n(\Omega X) & & & & & \\
 0 & 0 & \cdots & 0 & 0 & \\
 \vdots & \vdots & \ddots & \vdots & \vdots & \\
 0 & 0 & \cdots & 0 & 0 & \\
 \mathbb{Z} & 0 & \cdots & 0 & H_n(X) & H_{n+1}(X)
 \end{array}$$

where rows 1 through $n-1$ are all zeroes because the coefficient group $H_k(\Omega X)$ is zero for $1 \leq k < n$. Since the entire diagonal $p+q = n-2$ consists of zeroes, $H_n(X)$ can only ever map into zero, so it persists in the E^∞ page; since PX is contractible, this means $H_n(X)$ must be zero. To deal with $H_{n+1}(X)$, we note that the entry above $H_n(X)$ is zero by hypothesis if $n > 1$, and if $n = 1$ then this entry sits directly to the right of $\pi_n(X)$, so there is no differential from it into $\pi_n(X)$. Thus the only nontrivial differential involving $H_{n+1}(X)$ or $\pi_n(X)$ is the one in page $n+1$, where $H_{n+1}(X)$ maps into $\pi_n(X)$; this is the only opportunity for these two to become zero (which they must in E^∞), which means that

$$0 \rightarrow H_{n+1}(X) \xrightarrow{d^{n+1}} \pi_n(\Omega X) \rightarrow 0$$

is exact, i.e. $H_{n+1}(X) \cong \pi_n(\Omega X) \cong \pi_{n+1}(X)$. □

Example 2.7. Let $1 \rightarrow A \rightarrow B \rightarrow C \rightarrow 1$ be a short exact sequence of groups. Each homomorphism $B \rightarrow C$ can be realized by a map $K(B, 1) \rightarrow K(C, 1)$. Converting this into a fibration and feeding it into the long exact sequence shows that

$$1 \rightarrow \pi_1(F) \rightarrow B \rightarrow C \rightarrow 1$$

is exact, so $\pi_1(F) = A$.

elaborate

uh ok this is more complicated than i thought

2.3 Atiyah-Hirzebruch spectral sequence

TODO

a-h ss

2.4 Adams spectral sequence

TODO

adams ss

3 Practice

Exercise 3.1. Show that for a sphere bundle $S^n \hookrightarrow E \xrightarrow{\pi} B$ with B simply connected, there is an exact sequence of cohomology groups, called the **Gysin sequence**.

seq

Moreover, show that the map is obtained by cupping with a particular degree $n + 1$ cohomology class only depending on E (this is the **Euler class** of the vector bundle $E' \rightarrow B$ created by removing all the points at infinity of the fibers, thinking of each fiber as a compactified $\mathbb{R}^n$). Show also that the map is equal to π^* .

map

is this kosher?

Exercise 3.2. What does the cohomology Serre spectral sequence say about a Serre fibration with base space S^n for $n > 1$? (You should get a long exact sequence that looks like the Gysin sequence, called the **Wang sequence**.)

map

The following exercises are from Ajay.

Exercise 3.3. Use the Atiyah-Hirzebruch spectral sequence to compute the K -theory of $\mathbb{C}P^\infty$.

Exercise 3.4. Compute the homology of $K(\mathbb{Z}, 2) = \mathbb{C}P^\infty$ using the Serre spectral sequence, noting that there is a fibration $S^\infty \rightarrow \mathbb{C}P^\infty$ (what is its fiber?). Note that you should be able to verify your answer easily using cellular homology.

Exercise 3.5. Do the same as the above exercise for $\mathbb{R}P^\infty$ (perhaps with $\mathbb{Z}_2$ coefficients; I am not sure how tractable this is).

Exercise 3.6. If $E \rightarrow B$ is a covering space, how are the homology of E and B related?

Exercise 3.7. Deduce the Künneth formula for (co)homology with coefficients in a field using the Serre spectral sequence.

Bonus question: What about the case where the coefficient ring is not a field?

Exercise 3.8. Compute the cohomology of ΩS^n , the loop space of the n -sphere. What is its cup product structure?

Exercise 3.9. If $E \rightarrow B$ is a fibration whose fiber is contractible, what can be said about the homology of E and B ? (For a spoiler of the answer, feed the fibration into the homotopy long exact sequence.) What about if B is contractible, or E is contractible?

Exercise 3.10. For each positive integer m , let $\mathrm{SL}_2(\mathbb{Z})[m]$ be the subgroup of $\mathrm{SL}_2(\mathbb{Z})$ given by matrices congruent to identity modulo m . $\mathrm{SL}_2(\mathbb{Z})[m]$ is a finite-index normal subgroup of $\mathrm{SL}_2(\mathbb{Z})$. In fact, it is free for $m \geq 3$.

Let V be a $\mathbb{Q}$ -vector space. Show that

$$H^k(\mathrm{SL}_2(\mathbb{Z}); V) \cong H^k(\mathrm{SL}_2(\mathbb{Z})[m]; V)^{\mathrm{SL}_2(\mathbb{Z}/m\mathbb{Z})}.$$

Conclude that

$$H^0(\mathrm{SL}_2(\mathbb{Z}); \mathbb{Z}) \cong \mathbb{Z}, \quad H^1(\mathrm{SL}_2(\mathbb{Z}); \mathbb{Z}) \cong 0, \quad H^2(\mathrm{SL}_2(\mathbb{Z}); \mathbb{Z}) \cong \mathbb{Z}/12\mathbb{Z}$$

and that $H^k(\mathrm{SL}_2(\mathbb{Z}); \mathbb{Z})$ is a finitely generated torsion group for all $k \geq 2$. Some buzzwords that may be helpful as you work through this challenge: the Lyndon-Hochschild-Serre spectral sequence; Maschke's theorem.

(If you do work this out, congrats!! You've computed the zeroth, first, and second cohomology groups of the moduli space of elliptic curves, and also worked out that the higher cohomology groups are all finitely generated torsion groups.)

Some more, from Nick:

Exercise 3.11. Prove the various universal coefficient theorems.

Exercise 3.12. Show that Ext and Tor are well-defined. That is, the two right derived functors which define $\text{Ext}_R^n(A, B)$ agree and the two left derived functors which define $\text{Tor}_n^R(A, B)$ agree.

more purely algebraic exercises, as well as a-h ss and adams ss exercises